Article

On the minimum dataset size and quality for a maize seedling object detector

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Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

In the last decade, the use of object detectors in agricultural applications has gained significant attention. Object detectors are algorithms designed to identify and locate objects within an image. They are widely used in precision agriculture for tasks such as crop monitoring, yield estimation, and disease detection. Here we focus on the task of counting maize seedlings in open field.

Plant counting is a critical operation in precision agriculture, plant breeding, and products evaluation. Accurate plant counts can provide valuable information for crop management, yield estimation, and disease detection to both farmers and researchers. This task was often performed manually by human operators. Now a days it is the more and more common to use computer vision algorithms to automate this process.

The development of these algorithms has evolved from traditional methods, such as sliding window approaches and handcrafted features, to modern deep learning-based techniques for improved accuracy and efficiency.

One of the critical challenges in training effective object detectors is the availability of a sufficiently large and high-quality dataset. This study aims to determine the minimum dataset size and quality required to train a robust object detector for identifying maize seedlings in georeferenced orthomosaics. Georeferenced orthomosaics are images created through photogrammetry, a process that involves capturing georeferenced overlapping aerial images and stitching them together to form a single, seamless image.

The modern approach to object detection involves the use of deep learning models, such as Convolutional Neural Networks (CNNs) and Transformer-based models. CNNs,

Received:
Revised:
Accepted:
Published:

Citation: Lastname, F.; Lastname, F.; Lastname, F. Title. *Journal Not Specified* **2025**, *1*, 0. https://doi.org/

including architectures like ResNet, Darknet, and EfficientNet, have been widely used for their ability to capture spatial hierarchies in images. Transformer-based models, such as DETR (Detection Transformer) and ViT (Vision Transformer), have gained attention in the last years for their capability to model long-range dependencies and achieve state-of-the-art performance in various vision tasks. Zero-shot and few-shot object detection are two emerging paradigms in object detection. Models in this category aim to detect classes with no or very few labeled examples. They often leverage feature transfer or meta-learning components to generalize across classes under extreme data scarcity. This approach reduces the annotation burden and is especially beneficial in domains where collecting exhaustive training data is impractical. Zero-shot object detection detects new categories without any training samples by leveraging semantic relationships or contextual information learned from known classes. Few-shot approaches optimize models to learn quickly from a handful of labeled examples. Some prominent zero-shot models include LSDA, ViLD, and ZSD; among the few-shot methods, TFA, FSCE, and Meta R-CNN are commonly cited for their strong performance under data-scarce conditions.

Training a classic deeplearning object detector requires a large dataset of annotated images, where each image is labeled with the location and class of the objects of interest. The process of annotating images can be time-consuming and labor-intensive, as it often involves manual labeling by human annotators. The quality of the dataset, including factors such as annotation accuracy, diversity of samples, and class balance, plays a crucial role in the performance of the object detector. Zero-shot and few-shot object detection models, beeing designed to alleviate the need for large annotated datasets could be the solution to the problem of limited data availability.

In this study, we investigate the minimum number of images and annotations required to train an effective maize seedling object detector. By determining the optimal dataset size and quality, we aim to provide guidelines for researchers and practitioners in the field of precision agriculture, enabling them to develop efficient and accurate object detection models with limited resources.

2. Materials and Methods

The benchmarking of object detectors is typically performed on standard datasets, while for the use-case of maize seedling count, international standards can be used .

3. Results

This section may be divided by subheadings. It should provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn.

3.1. Subsection

3.1.1. Subsubsection

Bulleted lists look like this:

- First bullet;
- Second bullet;
- Third bullet.

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The text continues here.

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All figures and tables should be cited in the main text as Figure 1, Table 1, etc.

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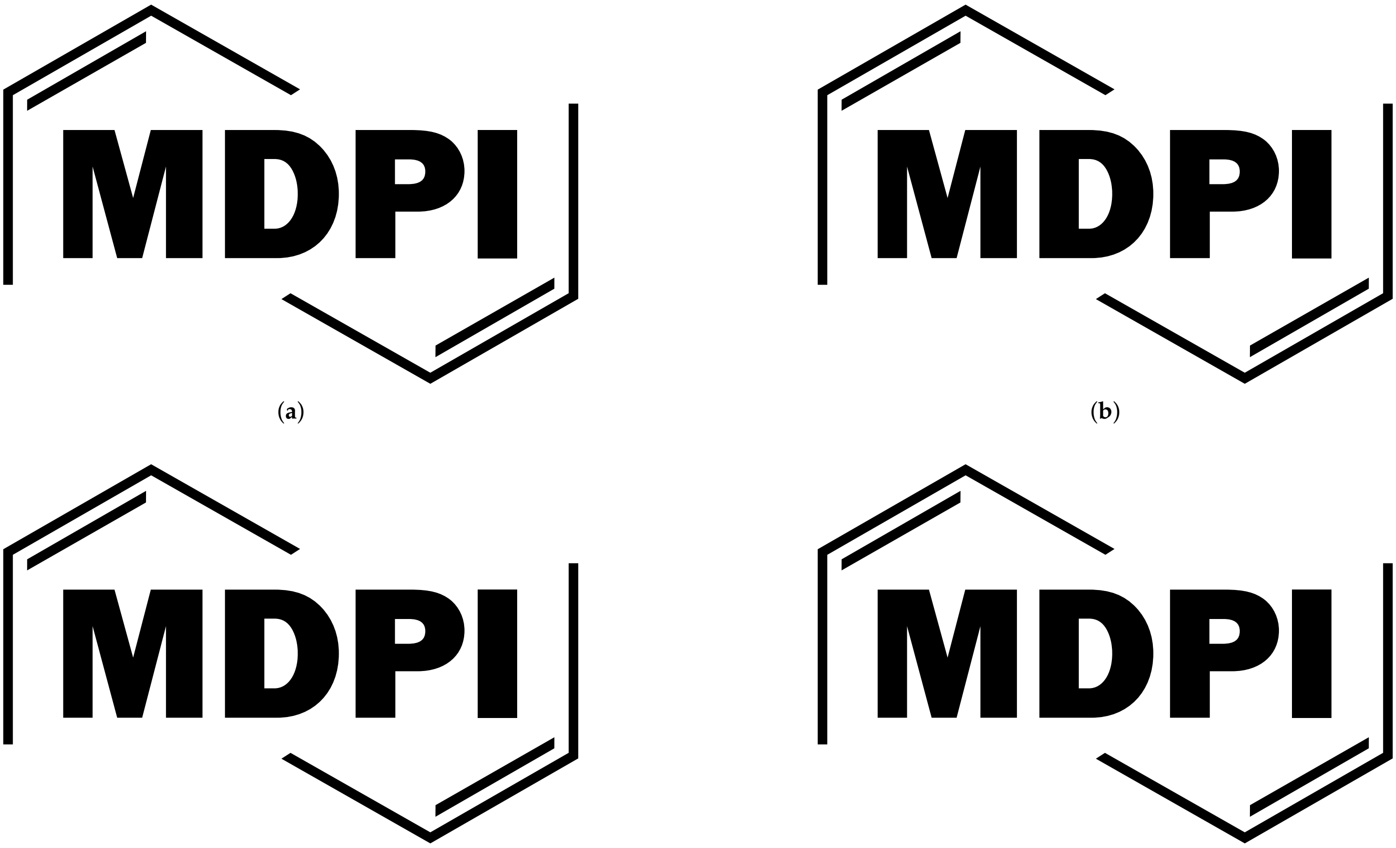
Title 1	**Title 2**	**Title 3**
Entry 1	Data	Data
Entry 2	Data	Data [1]

[1] Tables may have a footer.

The text continues here (Figure 2 and Table 2).



(**a**) (**b**) (**c**) (**d**)

Figure 2. This is a wide figure. Schemes follow the same formatting. If there are multiple panels, they should be listed as: (**a**) Description of what is contained in the first panel. (**b**) Description of what is contained in the second panel. (**c**) Description of what is contained in the third panel. (**d**) Description of what is contained in the fourth panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.

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Entry 2	Data	Data	Data
	Data	Data	Data
	Data	Data	Data

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Text.

Text.

3.3. Formatting of Mathematical Components

This is the example 1 of equation:

$$a = 1, \tag{1}$$

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

This is the example 2 of equation:

$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z \tag{2}$$

Please punctuate equations as regular text. Theorem-type environments (including propositions, lemmas, corollaries etc.) can be formatted as follows:

Theorem 1. *Example text of a theorem.*

The text continues here. Proofs must be formatted as follows:

Proof of Theorem 1. Text of the proof. Note that the phrase "of Theorem 1" is optional if it is clear which theorem is being referred to. □

The text continues here.

4. Discussion

Authors should discuss the results and how they can be interpreted from the perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible. Future research directions may also be highlighted.

5. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

6. Patents

This section is not mandatory, but may be added if there are patents resulting from the work reported in this manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

Table A1. This is a table caption.

Title 1	Title 2	Title 3
Entry 1	Data	Data
Entry 2	Data	Data

Appendix B

All appendix sections must be cited in the main text. In the appendices, Figures, Tables, etc. should be labeled, starting with "A"—e.g., Figure A1, Figure A2, etc.

References

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3. Author 1, A.; Author 2, B. *Book Title*, 3rd ed.; Publisher: Publisher Location, Country, 2008; pp. 154–196.
4. Author 1, A.B.; Author 2, C. Title of Unpublished Work. *Abbreviated Journal Name* year, *phrase indicating stage of publication (submitted; accepted; in press)*.
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